

# Response to Lab 10 Feedback

I carefully reviewed the Lab 10 peer feedback and the later instructor feedback on the final project. The most important revision was a change in the role of the empirical component. In the earlier version, the project was framed as a logit based empirical evaluation of deceptive alignment. After revision, the final paper is now positioned as a review paper on sycophancy evaluation in preference aligned LLMs. The logit based diagnostic is no longer presented as the central contribution of the paper; it is kept only as supplementary exploratory code in the GitHub repository.

## Major revisions made in response to feedback:

### 1. Clarified the final paper's scope

Earlier versions mixed a literature review with a small empirical experiment. This created a weak connection between the review taxonomy and the logit based FPR results. In the final revision, I removed the experimental section from the main paper and focused the paper on review methodology, prompt design, metric selection, and evaluation gaps in sycophancy research.

The final paper now focuses on the following question: How do existing studies define, elicit, and measure sycophantic behavior in preference aligned LLMs, and where do current, metrics fail?

### 2. Strengthened the review methodology

The original method section was too general and looked only superficially systematic. I revised it into a PRISMA guided review structure. The paper now includes:

1. search sources
2. database level retrieval counts
3. representative search queries
4. inclusion and exclusion criteria
5. a PRISMA style screening flow
6. coding dimensions for the included studies

This makes the review process more transparent and easier to verify.

### 3. Added evidence from the coded papers

The previous taxonomy read too much like a conceptual classification. To make the Results section more empirical, I added coded distribution tables showing how the 48 included studies were grouped by:

1. evaluation format
2. prompt design
3. metric type

This makes the taxonomy more clearly derived from the reviewed studies rather than only from my own conceptual grouping.

### 4. Rewrote the paper's main argument

The revised paper now centers on prompt and metric mismatch. The key point is that different studies use similar terms such as “sycophancy”, “truthfulness failure” or “reward hacking”, but they often measure different behaviors.

For example:

1. agreement rate works for direct false agreement but fails on hedged correction.
2. truthfulness accuracy can detect false answers but may miss whether the model corrected the user's false premise.
3. LLM as a judge scoring can reward polished or polite responses even when the model avoids clear factual correction.
4. Reward model scores are useful for reward hacking analysis, but do not directly measure conversational sycophancy.

### 5. Revised the writing style

Based on feedback about AI like phrasing, I rewrote the Abstract, Introduction, and Discussion to reduce generic review paper language. The revised version uses more specific claims about why particular metrics fail, instead of relying on broad statements such as “metric mismatch is a problem.”

### 6. Kept the logit based experiment as supplementary material only

The earlier codebase included a logit based diagnostic using TruthfulQA style binary choice prompts. I kept this component in the GitHub repository as supplementary material because it still illustrates one controlled way to measure misleading option

selection.

However, I no longer present it as the main contribution of the final paper. The README now explains that the diagnostic is exploratory and should not be interpreted as a full benchmark or as proof that one alignment method is generally safer than another.

## 7. Updated the GitHub repository organization

I reorganized the GitHub repository so that it better matches the final project:

1. Paper/ contains the final paper and references.
2. extract\_logit\_lab\_9/ contains the supplementary logit based diagnostic.
3. Supplementary\_lab\_8/ contains earlier CNN course demonstration files.
4. README.md now explains that the final project is primarily a review paper, not an empirical benchmark paper.

I also clarified the role of the supplementary diagnostic so that it does not conflict with the revised paper.